

CANDIDATE
NAME

CENTRE
NUMBER

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CANDIDATE
NUMBER

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PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

October/November 2019

1 hour 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

- 1 When a light plastic ball is placed in a vertical column of moving air, the ball becomes stationary at a height h , as shown in Fig. 1.1.

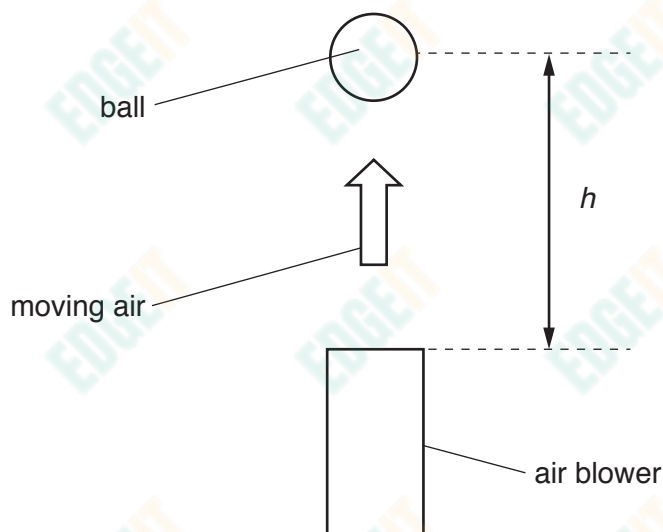


Fig. 1.1

A student is using an air blower to create the vertical column of moving air. The student connects the motor of the air blower to a d.c. power supply.

It is suggested that the relationship between the radius r of the ball and h is

$$\frac{4\pi r^3 gh}{3} = PK$$

where g is the acceleration of free fall, P is the power of the motor and K is a constant.

Design a laboratory experiment to test the relationship between r and h . Explain how your results could be used to determine a value for K .

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

Diagram

[illegible]

- 2 A student is investigating the oscillations of a mass attached to two springs connected in series, as shown in Fig. 2.1.

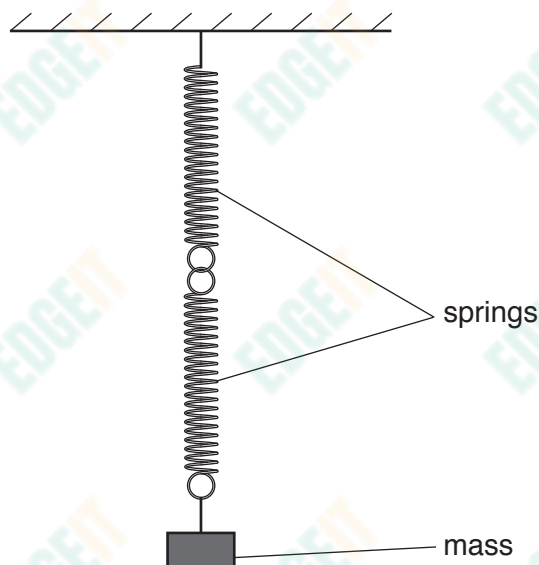


Fig. 2.1

A stopwatch is used to measure the time t for 10 oscillations. The measurement of t is repeated and the average period T is determined.

The experiment is repeated for different masses.

It is suggested that T and mass M are related by the equation

$$T = \frac{2\pi M^q}{\sqrt{k}}$$

where k is the spring constant of the two springs in series and q is a constant.

- (a) A graph is plotted of $\lg T$ on the y -axis against $\lg M$ on the x -axis.

Determine expressions for the gradient and the y -intercept.

gradient =

y -intercept =

[1]

(b) Values of M , $\lg(M/\text{g})$ and measurements of t are given in Fig. 2.2.

M/g	t/s	t/s	T/s	$\lg(M/\text{g})$	$\lg(T/\text{s})$
155	15.2	16.0		2.190	
205	18.3	17.5		2.312	
250	19.3	20.1		2.398	
305	21.0	21.8		2.484	
355	23.5	22.7		2.550	
410	24.1	24.9		2.613	

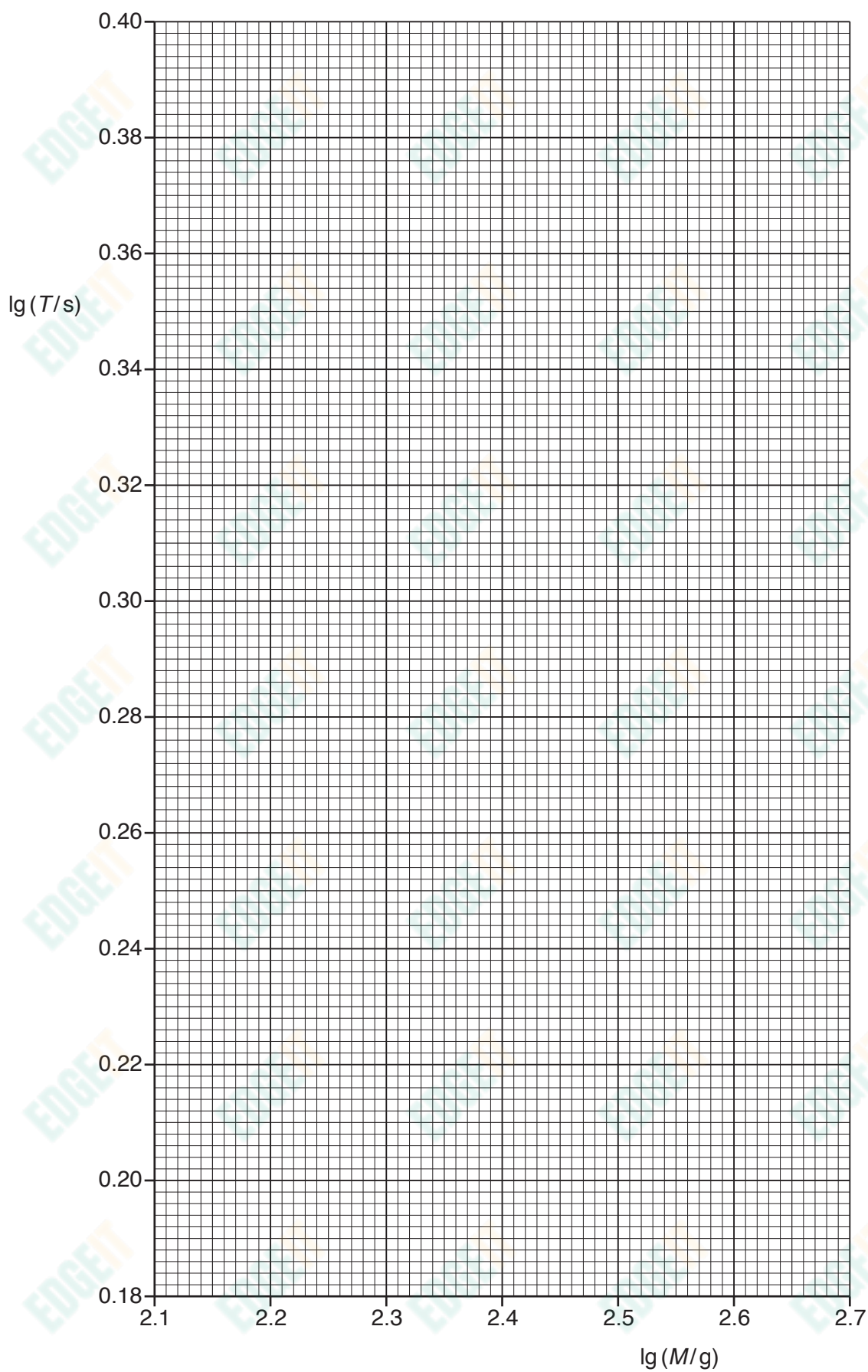
Fig. 2.2

Calculate and record values of T/s and $\lg(T/\text{s})$ in Fig. 2.2.
Include the absolute uncertainties in T/s and $\lg(T/\text{s})$.

[4]

- (c) (i) Plot a graph of $\lg(T/\text{s})$ against $\lg(M/\text{g})$.
Include error bars for $\lg(T/\text{s})$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled. [2]
- (iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the line of best fit. Do **not** include the absolute uncertainty.

y -intercept = [1]

- (d) Using your answers to (a), (c)(iii) and (c)(iv), determine the values of k and q . You need not be concerned with units. Do **not** include the absolute uncertainties.

k =

q =

[2]

- (e) Using your answers to (d), determine the mass M needed to give a period of 1.0s.

M =g [1]

[Total: 15]

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